

Marketing Homework 3

```
library(survival)
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

# preparing data
telco <- read.csv("telco.csv") %>%
  mutate(across(c(marital, ed, retire, gender, voice, internet, forward, churn), as.factor)) %>%
  mutate(churn_numeric = as.numeric(churn == "Yes"))

formula <- Surv(tenure, churn_numeric) ~ age + income + marital + ed + internet + voice

# Fit 4 AFT models
aft_weibull <- survreg(formula, data = telco, dist = "weibull")
aft_exp <- survreg(formula, data = telco, dist = "exponential")
aft_llogis <- survreg(formula, data = telco, dist = "loglogistic")
aft_lnorm <- survreg(formula, data = telco, dist = "lognormal")

# Compare AIC
model_comparison <- data.frame(
  Model = c("Weibull", "Exponential", "Log-Logistic", "Log-Normal"),
  AIC = c(AIC(aft_weibull), AIC(aft_exp), AIC(aft_llogis), AIC(aft_lnorm))
)

print(model_comparison)

##           Model           AIC
## 1      Weibull 3018.896
## 2 Exponential 3020.366
## 3 Log-Logistic 3011.891
## 4   Log-Normal 3012.683
```

```

#representative customer profile
newdata <- data.frame(
  age = 40,
  income = 60,
  marital = factor("Married", levels = levels(telco$marital)),
  ed = factor("College degree", levels = levels(telco$ed)),
  internet = factor("Yes", levels = levels(telco$internet)),
  voice = factor("Yes", levels = levels(telco$voice))
)

# Function to predict survival probabilities
predict_survival <- function(model, newdata, times, dist) {
  lp <- predict(model, newdata = newdata, type = "lp")
  scale <- model$scale

  if (dist == "weibull") {
    1 - pweibull(times, shape = 1/scale, scale = exp(lp))
  } else if (dist == "exponential") {
    1 - pexp(times, rate = 1/exp(lp))
  } else if (dist == "loglogistic") {
    1 / (1 + (times/exp(lp))^(1/scale))
  } else if (dist == "lognormal") {
    1 - plnorm(times, meanlog = lp, sdlog = scale)
  }
}

# survival curves
times <- seq(0, max(telco$tenure), length.out = 100)
surv_data <- data.frame(
  time = rep(times, 4),
  survival = c(
    predict_survival(aft_weibull, newdata, times, "weibull"),
    predict_survival(aft_exp, newdata, times, "exponential"),
    predict_survival(aft_llogis, newdata, times, "loglogistic"),
    predict_survival(aft_lnorm, newdata, times, "lognormal")),
  model = rep(c("Weibull", "Exponential", "Log-Logistic", "Log-Normal"), each = length(times))
)

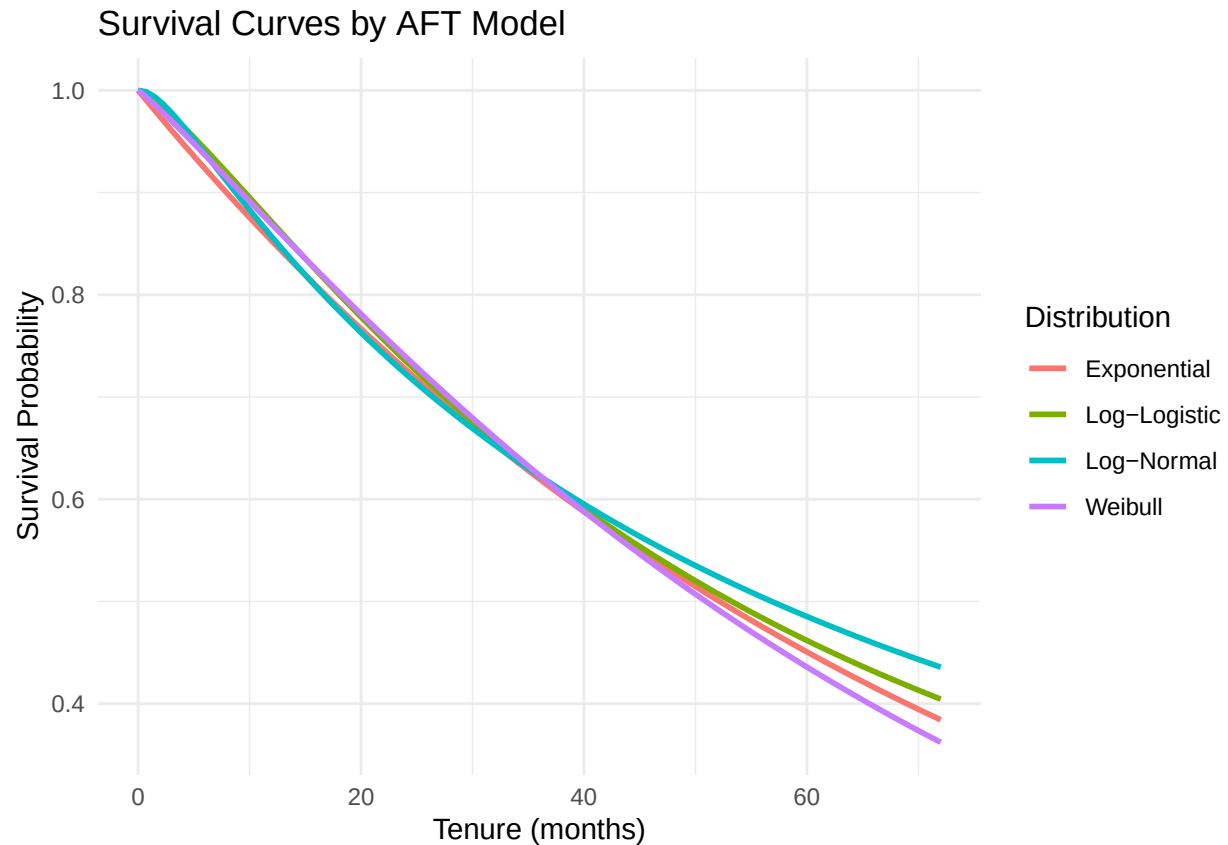
# Plot survival curves
ggplot(surv_data, aes(x = time, y = survival, color = model)) +
  geom_line(size = 1) +
  labs(title = "Survival Curves by AFT Model",
       x = "Tenure (months)",
       y = "Survival Probability",
       color = "Distribution") +
  theme_minimal()

```

```

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```



```
# Check significance of predictors
summary(aft_lnorm)
```

```
##
## Call:
## survreg(formula = formula, data = telco, dist = "lognormal")
##
```

	Value	Std. Error	z	p
## (Intercept)	2.474851	0.268834	9.21	< 2e-16
## age	0.057609	0.006169	9.34	< 2e-16
## income	0.001948	0.000997	1.95	0.051
## maritalUnmarried	-0.506081	0.124624	-4.06	4.9e-05
## edDid not complete high school	0.315536	0.216289	1.46	0.145
## edHigh school degree	0.291562	0.176910	1.65	0.099
## edPost-undergraduate degree	-0.014003	0.243555	-0.06	0.954
## edSome college	0.269203	0.179799	1.50	0.134
## internetYes	-0.681748	0.144493	-4.72	2.4e-06
## voiceYes	-0.173888	0.140968	-1.23	0.217
## Log(scale)	0.378066	0.046401	8.15	3.7e-16

```
##
## Scale= 1.46
##
## Log Normal distribution
## Loglik(model)= -1495.3   Loglik(intercept only)= -1602.5
##  Chisq= 214.35 on 9 degrees of freedom, p= 3.2e-41
## Number of Newton-Raphson Iterations: 4
## n= 1000
```

```
summary(aft_weibull)
```

```
##
## Call:
## survreg(formula = formula, data = telco, dist = "weibull")
##
##               Value Std. Error      z      p
## (Intercept)    2.939116   0.245094 11.99 < 2e-16
## age            0.050842   0.006097  8.34 < 2e-16
## income         0.001420   0.000987  1.44  0.1502
## maritalUnmarried -0.413027   0.111166 -3.72  0.0002
## edDid not complete high school 0.353144   0.206582  1.71  0.0874
## edHigh school degree 0.271264   0.156583  1.73  0.0832
## edPost-undergraduate degree 0.169024   0.199458  0.85  0.3968
## edSome college  0.210863   0.153062  1.38  0.1683
## internetYes     -0.632499   0.133511 -4.74 2.2e-06
## voiceYes        -0.162787   0.122266 -1.33  0.1831
## Log(scale)      -0.098699   0.051873 -1.90  0.0571
##
## Scale= 0.906
##
## Weibull distribution
## Loglik(model)= -1498.4   Loglik(intercept only)= -1606.4
##  Chisq= 215.96 on 9 degrees of freedom, p= 1.5e-41
## Number of Newton-Raphson Iterations: 7
## n= 1000
```

```
summary(aft_exp)
```

```
##
## Call:
## survreg(formula = formula, data = telco, dist = "exponential")
##
##               Value Std. Error      z      p
## (Intercept)    2.93334   0.27017 10.86 < 2e-16
## age            0.05422   0.00643  8.44 < 2e-16
## income         0.00151   0.00108  1.40 0.16019
## maritalUnmarried -0.43678   0.12188 -3.58 0.00034
## edDid not complete high school 0.38504   0.22744  1.69 0.09046
## edHigh school degree 0.29677   0.17232  1.72 0.08503
## edPost-undergraduate degree 0.16716   0.21986  0.76 0.44706
## edSome college  0.23085   0.16865  1.37 0.17106
## internetYes     -0.68862   0.14404 -4.78 1.7e-06
## voiceYes        -0.18339   0.13444 -1.36 0.17253
##
## Scale fixed at 1
##
## Exponential distribution
## Loglik(model)= -1500.2   Loglik(intercept only)= -1607
##  Chisq= 213.59 on 9 degrees of freedom, p= 4.6e-41
## Number of Newton-Raphson Iterations: 5
## n= 1000
```

```
summary(aft_llogis)
```

```
##
## Call:
## survreg(formula = formula, data = telco, dist = "loglogistic")
##
##              Value Std. Error      z      p
## (Intercept)    2.537355   0.255228  9.94 < 2e-16
## age            0.054581   0.006118  8.92 < 2e-16
## income         0.001534   0.000953  1.61  0.108
## maritalUnmarried -0.503528   0.118668 -4.24 2.2e-05
## edDid not complete high school 0.346215   0.211847  1.63  0.102
## edHigh school degree 0.283985   0.166566  1.70  0.088
## edPost-undergraduate degree -0.004995   0.227923 -0.02  0.983
## edSome college   0.212586   0.167134  1.27  0.203
## internetYes     -0.657199   0.137727 -4.77 1.8e-06
## voiceYes        -0.180466   0.131945 -1.37  0.171
## Log(scale)      -0.248759   0.051553 -4.83 1.4e-06
##
## Scale= 0.78
##
## Log logistic distribution
## Loglik(model)= -1494.9   Loglik(intercept only)= -1605.2
## Chisq= 220.52 on 9 degrees of freedom, p= 1.6e-42
## Number of Newton-Raphson Iterations: 4
## n= 1000
```

```
# log-logistic model with only significant predictors
final_formula <- Surv(tenure, churn_numeric) ~ age + marital + internet
final_model <- survreg(final_formula, data = telco, dist = "loglogistic")

# Check final model summary
summary(final_model)
```

```
##
## Call:
## survreg(formula = final_formula, data = telco, dist = "loglogistic")
##
##              Value Std. Error      z      p
## (Intercept)    2.63765   0.22929 11.50 < 2e-16
## age            0.05956   0.00558 10.68 < 2e-16
## maritalUnmarried -0.48694   0.11825 -4.12 3.8e-05
## internetYes     -0.83234   0.12141 -6.86 7.1e-12
## Log(scale)      -0.24506   0.05155 -4.75 2.0e-06
##
## Scale= 0.783
##
## Log logistic distribution
## Loglik(model)= -1499.5   Loglik(intercept only)= -1605.2
## Chisq= 211.5 on 3 degrees of freedom, p= 1.4e-45
## Number of Newton-Raphson Iterations: 4
## n= 1000
```

```

# Verify AIC improvement
AIC(final_model)

## [1] 3008.913

predict_survival <- function(model, newdata, times, dist) {
  lp <- predict(model, newdata = newdata, type = "lp")
  scale <- model$scale

  if (dist == "weibull") {
    1 - pweibull(times, shape = 1/scale, scale = exp(lp))
  } else if (dist == "exponential") {
    1 - pexp(times, rate = 1/exp(lp))
  } else if (dist == "loglogistic") {
    1 / (1 + (times/exp(lp))^(1/scale))
  } else if (dist == "lognormal") {
    1 - plnorm(times, meanlog = lp, sdlog = scale)
  }
}

#Preparing data
telco <- telco %>%
  mutate(
    across(c(marital, ed, retire, gender, voice, internet, forward, custcat, churn), as.factor),
    churn_numeric = as.numeric(churn == "Yes"),
    monthly_revenue = case_when(
      custcat == "Basic" ~ 30,
      custcat == "Plus" ~ 50,
      custcat == "Premium" ~ 80,
      TRUE ~ 40
    ),
    monthly_cost = case_when(
      internet == "Yes" & voice == "Yes" ~ 25,
      internet == "Yes" ~ 20,
      voice == "Yes" ~ 15,
      TRUE ~ 10
    ),
    monthly_margin = monthly_revenue - monthly_cost
  )

# survival model
final_model <- survreg(
  Surv(tenure, churn_numeric) ~ age + marital + income + internet + voice + custcat,
  data = telco,
  dist = "loglogistic"
)

# CLV calculation parameters
time_horizon <- 60
time_periods <- 1:time_horizon
annual_discount_rate <- 0.10
monthly_discount_rate <- (1 + annual_discount_rate)^(1/12) - 1

```

```

# survival probability matrix
survival_matrix <- sapply(time_periods, function(t) {
  predict_survival(final_model, telco, t, dist = "loglogistic")
})

# discounting
discount_factors <- (1 + monthly_discount_rate)^-time_periods
discounted_survival <- sweep(survival_matrix, 2, discount_factors, `*`)

# Calculate CLV
if (exists("telco") && "monthly_margin" %in% names(telco)) {
  telco$CLV <- telco$monthly_margin * rowSums(discounted_survival, na.rm = TRUE)
} else {
  stop("Required columns not found in telco dataframe")
}

if ("CLV" %in% names(telco)) {
  print(summary(telco$CLV))

  # 9. Visualize
  ggplot(telco, aes(x = CLV)) +
    geom_histogram(fill = "steelblue", bins = 30) +
    scale_x_continuous(labels = scales::dollar) +
    labs(title = "Customer Lifetime Value Distribution",
         x = "CLV", y = "Number of Customers") +
    theme_minimal()
} else {
  message("CLV column creation failed. Please check the calculation steps.")
  print(head(telco))
}

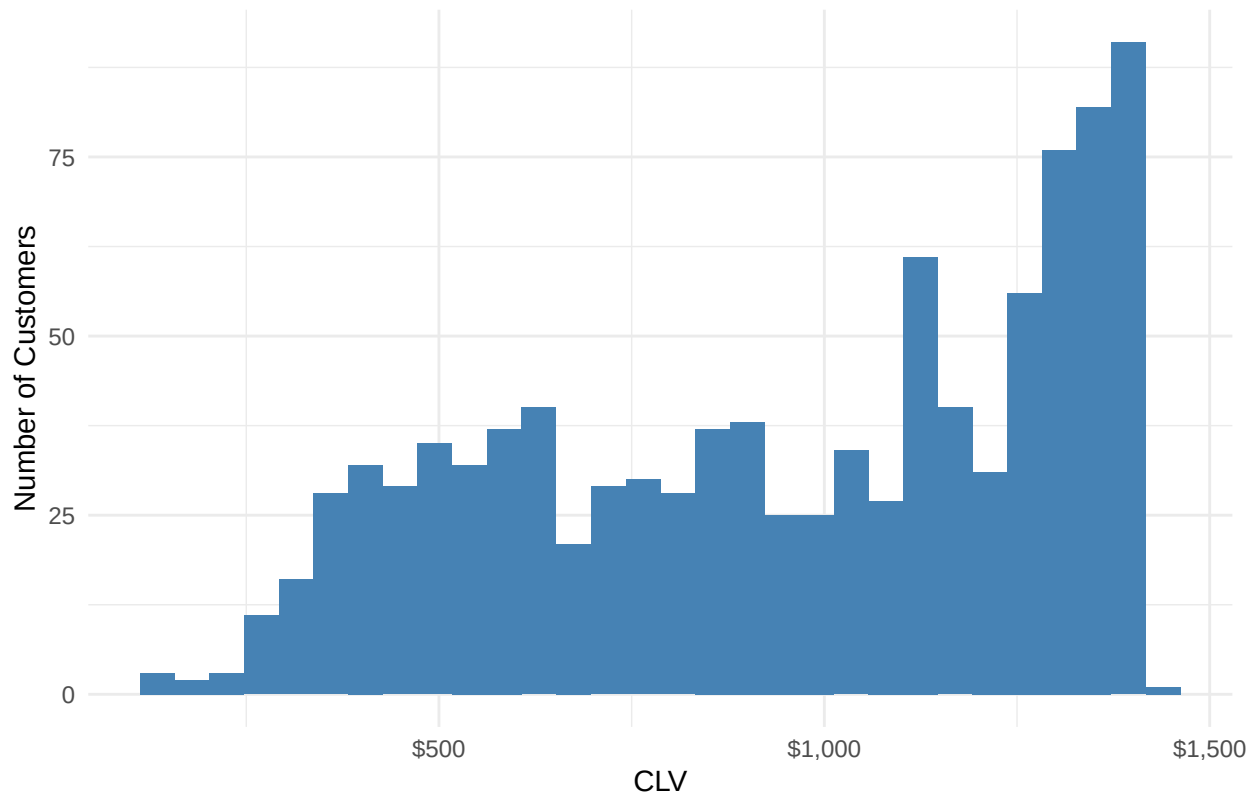
```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    113.5   633.7  1011.3   947.7  1281.1  1418.0

```

Customer Lifetime Value Distribution



```
# Segment analysis by gender and other demographics
```

```
gender_segments <- telco %>%
```

```
  group_by(gender) %>%
```

```
  summarise(
```

```
    avg_clv = mean(CLV, na.rm = TRUE),
```

```
    median_clv = median(CLV, na.rm = TRUE),
```

```
    customers = n(),
```

```
    churn_rate = mean(churn_numeric, na.rm = TRUE),
```

```
    avg_tenure = mean(tenure, na.rm = TRUE)
```

```
  ) %>%
```

```
  arrange(desc(avg_clv))
```

```
# gender segments
```

```
print(gender_segments)
```

```
## # A tibble: 2 x 6
```

```
##   gender avg_clv median_clv customers churn_rate avg_tenure
```

```
##   <fct>   <dbl>    <dbl>    <int>    <dbl>    <dbl>
```

```
## 1 Male     957.     1013.     483     0.271     35.0
```

```
## 2 Female   939.     1011.     517     0.277     36.0
```

```
demo_segments <- telco %>%
```

```
  group_by(
```

```
    gender,
```

```
    age_group = cut(age, breaks = c(18, 30, 40, 50, 60, 100),
```

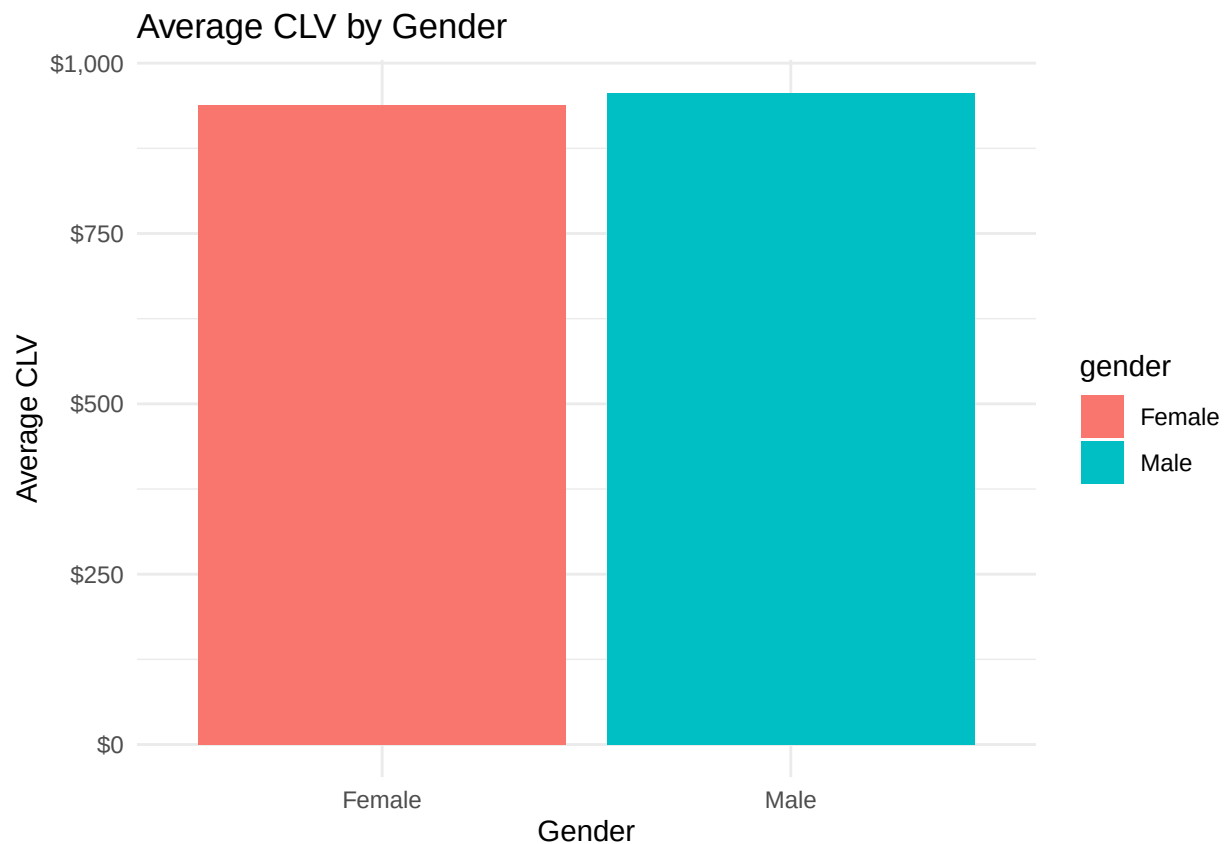


```

        labels = c("18-30", "31-40", "41-50", "51-60", "60+")),
  marital
) %>%
summarise(
  avg_clv = mean(CLV, na.rm = TRUE),
  customers = n(),
  .groups = 'drop'
) %>%
filter(customers > 10) %>% # Only show segments with sufficient samples
arrange(desc(avg_clv))

# gender differences
ggplot(gender_segments, aes(x = gender, y = avg_clv, fill = gender)) +
  geom_col() +
  scale_y_continuous(labels = scales::dollar) +
  labs(title = "Average CLV by Gender",
       x = "Gender",
       y = "Average CLV") +
  theme_minimal()

```

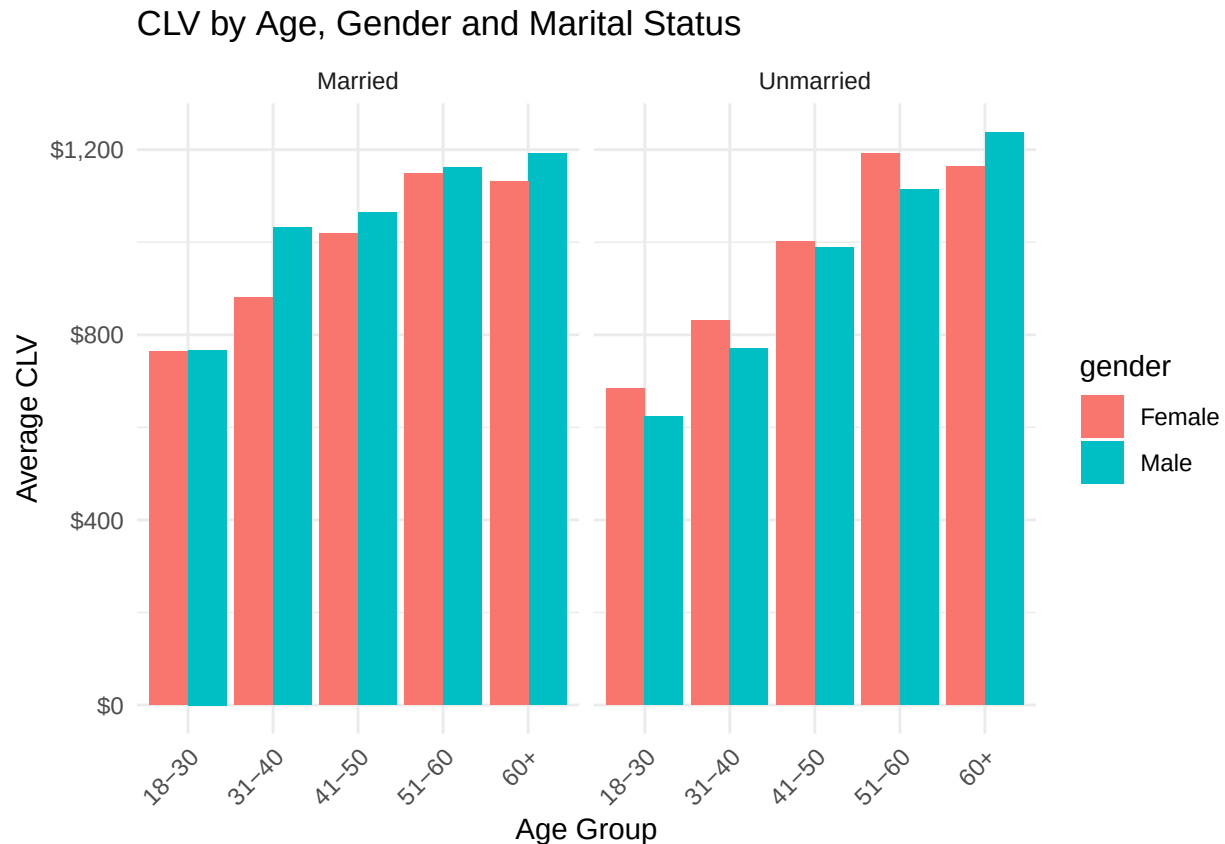


```

# age and gender patterns
ggplot(demo_segments, aes(x = age_group, y = avg_clv, fill = gender)) +
  geom_col(position = "dodge") +
  facet_wrap(~marital) +
  scale_y_continuous(labels = scales::dollar) +

```

```
labs(title = "CLV by Age, Gender and Marital Status",
     x = "Age Group",
     y = "Average CLV") +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Report

The results of my code showed that the Log-Logistic model has the lowest AIC (3011.891), indicating it provides the best balance of model fit and complexity for this dataset. This suggests customer churn follows a hazard rate that initially increases then decreases over time (characteristic of log-logistic distributions), which matches common customer lifecycle patterns where churn risk peaks after onboarding then declines for loyal customers. Also, I considered the summary info of each distribution. Hence, I chose it for my final model and eliminated insignificant variables.

From the CLV histogram we can say that the long tail on the left (lower CLV) shows that most customers cluster near higher values, but a minority of low-value customers drag the average down.

While calculating CLV of male and female customers separately, we can state that while males show slightly higher average CLV (+1.9%), the median values are nearly identical (1,013 vs 1,011), suggesting the small difference comes from outliers rather than central tendency. From the segmentation graphs we can say that CLV increases with the age and it is generally higher for married people. This makes total sense and aligns with real life scenarios. According to the graphs the most valuable segment is "Male" and "Married."

```
time_periods <- 1:120
monthly_discount_rate <- (1 + 0.07)^(1/12) - 1
```

```

telco <- telco %>%
  mutate(
    monthly_margin = monthly_revenue * 0.75 - monthly_cost
  )

telco <- telco %>%
  mutate(
    survival_1yr = predict_survival(
      final_model,
      newdata = telco,
      times = 12, # 1 year = 12 months
      dist = "loglogistic"
    )
  )

survival_matrix <- sapply(time_periods, function(t) {
  predict_survival(final_model, telco, t, dist = "loglogistic")
})
discounted_survival <- sweep(survival_matrix, 2, (1 + monthly_discount_rate)^-time_periods, `*`)
telco$CLV <- telco$monthly_margin * rowSums(discounted_survival, na.rm = TRUE)

at_risk <- telco %>%
  filter(
    survival_1yr < 0.7,
    CLV > 100,
    CLV < quantile(telco$CLV, 0.75, na.rm = TRUE)
  )

budget_params <- list(
  n_at_risk = nrow(at_risk),
  avg_clv_at_risk = mean(at_risk$CLV, na.rm = TRUE),
  total_clv_at_risk = sum(at_risk$CLV, na.rm = TRUE),
  retention_rate = 0.6,
  budget_pct = 0.15,
  suggested_budget = sum(at_risk$CLV, na.rm = TRUE) * 0.15 * 0.6
)

budget_breakdown <- at_risk %>%
  mutate(
    segment = case_when(
      CLV > quantile(telco$CLV, 0.6, na.rm = TRUE) ~ "High-Value",
      TRUE ~ "Standard"
    )
  ) %>%
  group_by(segment) %>%
  summarise(
    customers = n(),
    avg_clv = mean(CLV, na.rm = TRUE),

```

```

    budget_allocation = sum(CLV, na.rm = TRUE) * 0.15 * 0.6,
    .groups = "drop"
  )

cat("At-risk customers:", budget_params$n_at_risk, "\n")

## At-risk customers: 64

cat("Avg CLV at risk: $", round(budget_params$avg_clv, 2), "\n")

## Avg CLV at risk: $ 215.42

cat("Total CLV at risk: $", round(budget_params$total_clv_at_risk, 2), "\n")

## Total CLV at risk: $ 13786.99

cat("Suggested annual budget: $", round(budget_params$suggested_budget, 2), "\n\n")

## Suggested annual budget: $ 1240.83

print(budget_breakdown)

## # A tibble: 1 x 4
##   segment customers avg_clv budget_allocation
##   <chr>      <int>   <dbl>         <dbl>
## 1 Standard      64    215.         1241.

library(ggplot2)
ggplot(budget_breakdown, aes(x = segment, y = budget_allocation, fill = segment)) +
  geom_col() +
  geom_text(aes(label = paste0("$", round(budget_allocation, 0))), vjust = -0.3) +
  labs(title = "Recommended Retention Budget by Segment",
       subtitle = paste("Total Budget: $", round(budget_params$suggested_budget, 2)),
       x = "Customer Segment",
       y = "Budget Allocation") +
  scale_y_continuous(labels = scales::dollar) +
  theme_minimal()

```

Recommended Retention Budget by Segment

Total Budget: \$ 1240.83



The analysis reveals your Standard at-risk segment contains 64 customers with an average CLV of \$215, allocating \$1,241 for retention efforts (~\$19 per customer). This represents a low-value, high-churn population - likely basic plan subscribers or inactive accounts. While this budget supports cost-effective tactics like automated emails or minor incentives, the strikingly low CLV suggests potential issues:

Recommendations:

1. High-Value Win-Back Campaigns
 - Target: Customers who churned within last 3 months with top 25% CLV
 - Tactic: Offer 10-15% of their CLV as reactivation credit
2. At-Risk Customer Triage
 - Target: Customers with 40-60% 1-year survival probability
 - Tactic: Proactive retention offers before churn
3. Usage-Based Incentives
 - Target: Customers with >30% usage decline
 - Tactic: Data/voice boosters + "We noticed you're using less data. . ."
4. Contract Renewal Optimization
 - Target: Customers nearing contract end

- Tactic: Early renewal bonuses
- Execution: “Renew now and get [premium feature] free”

Key Metrics to Track: - Cost per Saved Customer (target <30% of CLV) - Retention Rate Lift (aim for +15-25%) - Saved CLV vs. Retention Spend